TU Clausthal

Requirements Engineering

Exercise – 4

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Deadline: 25.01.2027 – 1:59 PM

Submission location: Moodle Course Page

Task(s):

Submission guidelines:

- Please upload exactly ONE SNAKES python file, Simulation State PNG Images, and your AOM Goal and Behavior Interface models as PDF files
- Please use SNAKES CPN ONLY.
- Please review your submission once you have uploaded the CPN model and then click the Submit button.

Scenario:

In the Lecture we simulated the simple example of a Process reading a file and then writing back to it. Now, imagine the scenario, with two Processes trying to read or write to the same file.

Task(s):

1. Create a CPN model based on the AOM Goal and Behavioral Interface Models using the mapping heuristics discussed in the lecture.
2. The initial tokens (initial state) of your system should reflect at least 2 bikes and 2 commuters. More generally → Please ensure a level of complexity similar to the EV Charging CPN model example given in the lecture.
3. To make it easier for you, you can adapt and modify your solution from the Charging Station. Example here: https://github.com/ETCE-LAB/RE-CPN-Exercises
4. Since SNAKES CPN does not support hierarchical CPNs, you may omit this aspect.
5. You can choose a simple subset of the goals and only model those, **however:** this subset should include the *interesting* component of the system, i.e., the reserving, riding the E-Scooter, and computation of the ride cost. Optionally you can also include registration and payment process.
6. You can modify your AOM models if you feel the need to, as long as the updated model meets the task description from Exercise 4.

Questions? → etce-re@tu-clausthal.de

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